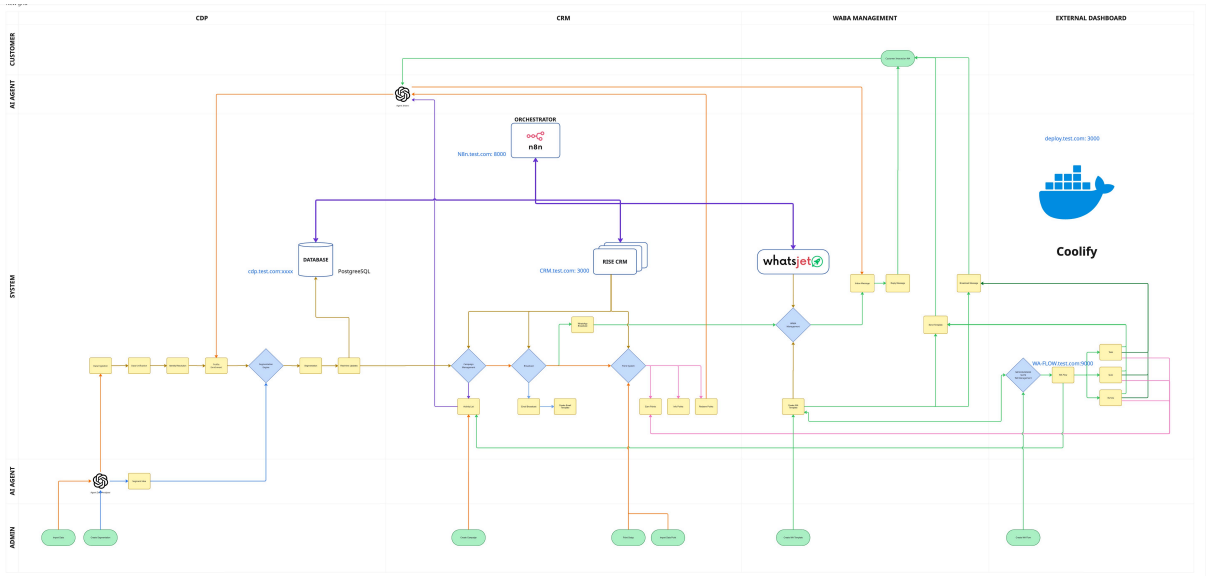
CDP CRM Platform Promina Indofood

Technical Development Roadmap

CDP + Rise CRM + WhatsApp + n8n Automation Platform



1. System Overview

Platform terdiri dari beberapa komponen utama:

1. CDP (Customer Data Platform)
2. CRM (Rise CRM + plugin customization)
3. Automation Orchestrator (n8n)
4. WhatsApp Channel (WABA + Flow)
5. Loyalty & Point Ledger
6. Consent & Suppression Layer
7. Analytics & Scoring

Tim Development:

- PIC CDP: 1 Developer
- PIC AI Agent: 1 Developer
- PIC CRM + n8n + Integration: 2 Developer

Total developer: 4

Target timeline: 8 minggu

Strategi development:

- Iterative delivery
- Foundation-first architecture
- Event-driven integration
- MVP → Production Hardening

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2. System Architecture Diagram

```
flowchart LR
    subgraph Channels[Customer Channels]
        WA[WhatsApp / WABA]
        WAF[WhatsApp Flow]
        EM[Email Provider]
        AG[AI Agent / Assistant]
    end

    subgraph ORCH[Integration & Orchestration]
        N8N[n8n Orchestrator]
        WEB[Webhook / Event Router]
        Q[Retry Queue / Job Handling]
    end

    subgraph CORE[Core Platform]
        CDP[CDP Core
Profile + Identity + Event Store + Segments]
        CRM[Rise CRM
Contacts + Campaigns + Activity + Plugins]
```

```
        CONS[Consent & Suppression Service]
        LED[Point Ledger Service
Earn / Burn / Expire / Redeem]
        SCORE[Scoring & Analytics]
    end

    subgraph DATA[Operational Data Sources]
        WABA[WABA / BSP Management
Raw Message Log + Delivery Status]
        DB[(Primary Database)]
        DW[(Analytics / Reporting Store)]
    end

    WA --> WABA
    WAF --> WABA
    WABA --> WEB
    EM --> WEB
    AG --> N8N

    WEB --> N8N
    N8N --> Q
    Q --> N8N

    CRM --> N8N
    N8N --> CRM

    N8N --> CONS
    CONS --> N8N

    N8N --> LED
    LED --> N8N

    N8N --> CDP
    CDP --> N8N

    WABA --> CRM
    WABA --> CDP
```

```
    CDP --> SCORE
    LED --> SCORE
    CRM --> SCORE
    SCORE --> DW

    CDP --> DB
    CRM --> DB
    CONS --> DB
    LED --> DB

    CDP -. segment / profile summary .-> CRM
    CDP -. audience / trigger .-> N8N
    CRM -. campaign config .-> N8N
    WABA -. sent / delivered / read / fail .-> CDP
    LED -. point balance / status .-> CRM
```

Penjelasan arsitektur

1. Channels

- WhatsApp / WABA: kanal outbound dan inbound utama.
- WhatsApp Flow: untuk survey, quiz, task, referral, dan point interaction.
- Email Provider: untuk email broadcast dan follow-up campaign.
- AI Agent: untuk assistance, routing, atau future conversational layer.

2. Integration & Orchestration

- n8n menjadi middleware utama untuk menjalankan workflow, routing, retry, scheduling, dan integrasi antar sistem.
- Webhook / Event Router menerima callback dari WABA, email provider, WA Flow, dan service lain.
- Retry Queue menangani failure, timeout, dan reprocessing.

3. Core Platform

- CDP Core menjadi pusat customer profile, identity resolution, event store, dan segment computation.

- Rise CRM menjadi operational layer untuk campaign management, contact view, activity list, dan plugin-based extension.
- Consent & Suppression Service memastikan opt-in, opt-out, blacklist, dan frequency control berjalan sebelum pengiriman campaign.
- Point Ledger Service menyimpan transaksi poin secara immutable untuk earn, burn, expire, dan redeem.
- Scoring & Analytics menghitung engagement score, campaign performance, dan activity insight.

4. Operational Data Sources

- WABA / BSP Management tetap menjadi tempat log teknis raw message dan delivery transport status.
- Primary Database menyimpan data operasional inti.
- Analytics / Reporting Store digunakan untuk dashboard, reporting, dan agregasi performa.

5. Alur data utama

1. Campaign dibuat di Rise CRM.
2. n8n mengambil konfigurasi campaign, melakukan consent check, lalu mengirimkan pesan ke WABA / Email Provider.
3. Customer menerima pesan dan dapat berinteraksi lewat WhatsApp Flow.
4. Semua event seperti sent, delivered, read, flow_started, flow_completed, redeem, dan referral masuk ke Webhook / Event Router.
5. Event diteruskan ke CDP untuk disimpan sebagai behavioral event.
6. CDP memperbarui profile, segment, dan summary operasional.
7. Summary yang dibutuhkan tim operasional disinkronkan kembali ke Rise CRM.
8. Jika event terkait loyalty, Point Ledger diperbarui dan balance terbaru dikirim ke CRM / WA notification.
9. Scoring & Analytics mengolah data untuk dashboard dan optimization.

6. Source of truth

- CDP: profile, event, segment, score input.
- Rise CRM: operational campaign, contact handling, activity summary.
- WABA / BSP: raw message transport log.
- Point Ledger: transaksi poin dan balance calculation.
- Consent Service: consent status dan suppression logic.

$1 Source of Truth:

CDP

- Customer profile
- Customer behavior event
- Segment membership

CRM (Rise CRM)

- Operational customer record
- Campaign management
- Marketing activity

WABA / BSP

- Raw message log
- Delivery status

Point Ledger

- Earn
- Burn
- Expire

n8n

- Workflow orchestration
- Integration layer

2A. Customer Data Flow Diagram

```
flowchart LR
    subgraph Sources[Data Sources]
        APP[App / Web / Form]
        WA[WhatsApp Interaction]
        FLOW[WhatsApp Flow]
        CRMU[Rise CRM User Input]
        EM[Email Interaction]
    end

    subgraph Intake[Intake Layer]
        WEB[Webhook Receiver]
        API[Event / Profile API]
        N8N[n8n Router]
    end

    subgraph CDPZONE[CDP Layer]
        IDR[Identity Resolution]
        PROF[Customer Profile]
        EV[Event Store]
        ATTR[Attribute Processor]
        SEG[Segmentation Engine]
    end

    subgraph OPS[Operational Systems]
        CRM[Rise CRM]
        CONS[Consent Service]
        SCORE[Scoring Engine]
    end

    APP --> API
    WA --> WEB
    FLOW --> WEB
    CRMU --> CRM
    EM --> WEB

    API --> N8N
    WEB --> N8N
    N8N --> IDR
```

```
    IDR --> PROF
    N8N --> EV
    EV --> ATTR
    PROF --> ATTR
    ATTR --> SEG
    ATTR --> SCORE

    CONS --> SEG
    SEG -. segment summary .-> CRM
    SCORE -. score summary .-> CRM
    PROF -. profile summary .-> CRM
    CRM -. profile enrichment / manual update .-> API
```

Penjelasan Customer Data Flow

Diagram ini menjelaskan bagaimana data customer dari berbagai channel masuk ke platform, diproses di CDP, lalu disinkronkan kembali ke CRM dalam bentuk summary operasional.

Alur utamanya:

1. Interaksi customer berasal dari app/web, WhatsApp, WhatsApp Flow, email, atau input operasional di CRM.
2. Semua data masuk melalui webhook atau API, lalu diteruskan ke n8n untuk routing awal.
3. CDP melakukan identity resolution agar banyak identifier tetap terhubung ke satu customer profile.
4. Event disimpan ke event store, sedangkan data profil diperkaya ke customer profile.
5. Attribute processor menghasilkan field turunan seperti last activity, engagement indicator, dan preference summary.
6. Segmentation engine dan scoring engine menghitung segment membership dan score.
7. Hasil ringkasan profil, segment, dan score dikirim ke Rise CRM untuk kebutuhan operasional tim.

2B. WA Campaign Flow Diagram

```
flowchart LR
    MKT[Marketing User in Rise CRM] --> CAMP[Create Campaig
n]
    CAMP --> AUD[Select Segment / Audience]
    AUD --> TEMP[Select WA Template / Flow]
    TEMP --> SCH[Schedule / Trigger Send]
    SCH --> N8N[n8n Orchestrator]

    N8N --> CONS[Consent & Suppression Check]
    CONS -->|eligible| WABA[WABA / BSP Send API]
    CONS -->|blocked| SUP[Suppressed / Hold List]

    WABA --> CUST[Customer]
    CUST -->|open / reply / click| INB[Inbound / Interactio
n Webhook]
    CUST -->|start flow| FLOW[WhatsApp Flow]
    FLOW -->|submit| WEB[Flow Submission Webhook]

    WABA -->|sent / delivered / read / fail| STAT[Message S
tatus Webhook]
    INB --> N8N
    WEB --> N8N
    STAT --> N8N

    N8N --> CDP[CDP Event Store]
    N8N --> CRMLOG[CRM Activity Update]
    N8N --> LED[Point Ledger / Reward Logic]

    LED --> N8N
    N8N --> NOTIF[Send Follow-up / Point Info / Redeem Inf
o]
```

Penjelasan WA Campaign Flow

Diagram ini menggambarkan flow campaign WhatsApp dari sisi operasional marketing sampai closed-loop event.

Alurnya:

1. Tim marketing membuat campaign di Rise CRM.
2. Marketing memilih segment, template, atau WhatsApp Flow yang akan dipakai.
3. Campaign diteruskan ke n8n untuk orkestrasi.
4. Sebelum pengiriman, n8n menjalankan consent check dan suppression filter.
5. Pesan yang lolos dikirim ke WABA / BSP.
6. Customer bisa membaca pesan, membalas, atau masuk ke WhatsApp Flow.
7. Semua status pesan dan hasil Flow kembali lewat webhook.
8. n8n mencatat event ke CDP, memperbarui CRM activity, dan memicu reward atau follow-up jika relevan.

2C. Point Ledger Architecture

```
flowchart LR
    subgraph Triggers[Point Triggers]
        QUIZ[Quiz Complete]
        SURV[Survey Submit]
        TASK[Task Complete]
        REF[Referral Success]
        RED[Redeem Request]
        EXP[Expiration Job]
        ADJ[Manual Adjustment]
    end

    subgraph Rules[Rule Layer]
        RULE[Point Rule Engine]
        VAL[Validation / Anti Fraud]
    end

    subgraph Ledger[Ledger Layer]
        TXN[Point Transaction Generator]
        LEDGER[(Point Ledger - Immutable)]
```

```
        BAL[Balance Calculator]
        REDEEM[(Redemption Record)]
    end

    subgraph Outputs[Outputs]
        CRM[Rise CRM Summary]
        CDP[CDP Event + Profile]
        WA[WA Notification / Info]
        REP[Analytics / Audit]
    end

    QUIZ --> RULE
    SURV --> RULE
    TASK --> RULE
    REF --> VAL
    VAL --> RULE
    RED --> VAL
    EXP --> RULE
    ADJ --> RULE

    RULE --> TXN
    TXN --> LEDGER
    LEDGER --> BAL
    RED --> REDEEM
    BAL --> CRM
    BAL --> CDP
    TXN --> CDP
    BAL --> WA
    LEDGER --> REP
    REDEEM --> REP
```

Penjelasan Point Ledger Architecture

Point ledger dipisahkan dari CRM agar seluruh transaksi poin memiliki audit trail yang jelas dan immutable.

Prinsipnya:

1. Semua aktivitas yang menghasilkan atau menggunakan poin masuk sebagai trigger.
2. Rule engine menentukan apakah transaksi earn, burn, expire, atau adjustment valid dijalankan.
3. Validation layer mengecek anti-fraud untuk referral, redeem, atau aktivitas yang sensitif.
4. Point transaction generator membuat transaksi ledger yang tidak boleh diedit sembarangan.
5. Balance dihitung dari akumulasi ledger, bukan dari input manual.
6. Summary balance dikirim ke CRM dan CDP, sedangkan detail transaksi tetap berada di ledger.

2D. Event Processing Pipeline

```
flowchart LR
    subgraph Input[Event Sources]
        WABA[WABA Status Webhook]
        FLOW[WA Flow Submission]
        CRM[CRM Action]
        APP[App / Web Event]
        LED[Point Ledger Event]
    end

    subgraph Processing[Processing Layer]
        WH[Webhook Gateway]
        N8N[n8n Event Router]
        VAL[Schema Validation]
        MAP[Event Normalization / Mapping]
        DEDUP[Deduplication / Idempotency]
        PRI[Priority Split
Real-time vs Batch]
    end

    subgraph Storage[Storage Layer]
        EV[(CDP Event Store)]
```

```
        RAW[(Raw Log / Audit Store)]
        BATCH[(Batch Aggregation Queue)]
    end

    subgraph Consumers[Consumers]
        PROF[Profile Updater]
        SEG[Segment Engine]
        SCORE[Scoring Engine]
        CRMU[CRM Summary Sync]
        ANA[Analytics / Dashboard]
        AUTO[Automation Trigger]
    end

    WABA --> WH
    FLOW --> WH
    CRM --> WH
    APP --> WH
    LED --> WH

    WH --> N8N
    N8N --> VAL
    VAL --> MAP
    MAP --> DEDUP
    DEDUP --> PRI

    PRI --> EV
    PRI --> RAW
    PRI --> BATCH

    EV --> PROF
    EV --> SEG
    EV --> SCORE
    EV --> AUTO
    BATCH --> ANA
    PROF --> CRMU
    SEG --> CRMU
    SCORE --> CRMU
```

Penjelasan Event Processing Pipeline

Pipeline ini memastikan semua event dari berbagai sistem diproses secara konsisten sebelum masuk ke CDP.

Tahapannya:

1. Event masuk dari WABA, WA Flow, CRM, app/web, dan point ledger.
2. Semua event diterima lewat webhook gateway dan diarahkan oleh n8n.
3. Event diperiksa struktur dan validitasnya melalui schema validation.
4. Event dinormalisasi agar semua sumber mengikuti format event yang seragam.
5. Sistem melakukan deduplication atau idempotency check agar event ganda tidak diproses dua kali.
6. Event dipisah antara kebutuhan real-time dan batch.
7. Event real-time masuk ke CDP event store untuk profile update, segment update, scoring, dan trigger automation.
8. Event batch digunakan untuk agregasi reporting, dashboard, dan analytics.

2E. Critical Path Diagram (Engineering Dependency Map)

```
flowchart LR

    SPEC[Architecture + Event Schema]

    CDP1[CDP Core
Customer Profile]
    CDP2[Identity Resolution]
    CDP3[Event Ingestion API]

    CRM1[Rise CRM Setup]
    CRM2[Contact Sync CDP → CRM]
    CRM3[Campaign Module]

    WA1[WABA Integration]
```

```
    WA2[Template + Broadcast]
    WA3[WA Flow Integration]

    AUTO1[n8n Workflow Infrastructure]
    AUTO2[Campaign Automation]

    LED1[Point Ledger Core]
    LED2[Reward / Redeem Logic]

    CONS1[Consent + Suppression]

    LOOP1[Closed Loop Tracking
Delivery / Read / Flow]

    SCORE1[Scoring Engine]
    DASH[Analytics Dashboard]


    SPEC --> CDP1
    CDP1 --> CDP2
    CDP2 --> CDP3

    SPEC --> CRM1
    CDP3 --> CRM2
    CRM1 --> CRM2
    CRM2 --> CRM3

    CRM3 --> WA1
    WA1 --> WA2
    WA2 --> WA3

    WA2 --> AUTO1
    AUTO1 --> AUTO2

    CDP3 --> LED1
    LED1 --> LED2

    WA2 --> CONS1
```

```
    CONS1 --> AUTO1

    WA2 --> LOOP1
    WA3 --> LOOP1

    LOOP1 --> SCORE1
    SCORE1 --> DASH
```

Penjelasan Critical Path

Diagram ini menunjukkan **dependency engineering** yang menentukan kecepatan development.

Critical path utama:

Architecture → CDP Core → Event Ingestion → CRM Sync → Campaign Module → WABA Integration → Automation → Closed Loop Tracking → Analytics

Jika salah satu komponen di jalur ini terlambat, seluruh sistem akan ikut tertunda.

Dependency penting:

1. **CDP Core harus selesai sebelum Contact Sync dan Event Processing.**
2. **CRM Campaign Module harus siap sebelum integrasi WABA.**
3. **WABA Integration harus selesai sebelum Automation Workflow.**
4. **Closed Loop Tracking harus ada sebelum Analytics dan Scoring berjalan.**

Workstream paralel yang memungkinkan timeline tetap cepat:

- CDP Development
- CRM Customization
- AI Agent Development
- Integration + n8n Workflow

Karena beberapa pekerjaan berjalan paralel, timeline total tetap dapat diselesaikan dalam **6–8 minggu**.

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3. DEVELOPMENT PHASES

Phase 0 – Technical Specification

Output:

- Event schema
- Data model
- API contract
- Identity resolution rules
- Segment rules definition
- Point rules
- Consent model

Duration: 3–4 hari

Phase 1 – CDP Foundation

Scope:

- Customer profile service
- Identity resolution
- Event ingestion API
- Event store
- Profile enrichment
- Basic segmentation engine

Key tables:

- customer_profile
- customer_identity
- event_store
- segment_definition

Deliverables:

- Event ingestion endpoint
- Real-time event pipeline
- Profile update logic

Owner: CDP Developer

Duration: Week 1–2

Phase 2 – CRM Operational Layer (Rise CRM)

Scope:

- Rise CRM setup
- Custom plugin development
- Contact sync from CDP
- Campaign management module
- Activity list
- Segment selection

Deliverables:

- CRM contact sync service
- Campaign creation UI
- Activity history panel

Owner: CRM Team

Duration: Week 2–3

Phase 3 – WhatsApp Integration

Scope:

- WABA integration
- Template management
- Broadcast messaging
- Inbox message handling
- WA Flow integration

Flow events:

- flow_started
- flow_completed
- survey_submitted

- quiz_completed

Deliverables:

- Send message API
- Message status webhook
- Flow submission webhook

Owner: CRM + Integration Team

Duration: Week 3–4

Phase 4 – n8n Automation Layer

Scope:

- Workflow automation
- Campaign orchestration
- Retry logic
- Webhook processing

Example workflows:

- campaign_send
- flow_complete → point earn
- redeem_request → validation

Deliverables:

- Workflow library
- Retry queue
- Error monitoring

Owner: Integration Team

Duration: Week 4–5

Phase 5 – Loyalty & Point Ledger

Scope:

- Point earn
- Point burn

- Expiration rules
- Ledger transaction model

Key tables:

- point_ledger
- point_balance
- redemption

Deliverables:

- Immutable transaction ledger
- Balance calculation service

Owner: CDP Developer

Duration: Week 5–6

Phase 6 – Consent & Suppression

Scope:

- Opt-in capture
- Opt-out processing
- Suppression logic

Key tables:

- consent_record
- suppression_list

Deliverables:

- Consent API
- Suppression filter before broadcast

Owner: CDP + CRM Team

Duration: Week 6

Phase 7 – Closed Loop Event Tracking

Scope:

- Message delivery tracking

- Read status tracking
- Campaign outcome tracking

Events:

- message_sent
- message_delivered
- message_read

Deliverables:

- Event collector
- CDP event writeback

Owner: CDP + Integration

Duration: Week 6–7

Phase 8 – Scoring & Analytics

Scope:

- Engagement scoring
- Customer activity scoring
- Campaign dashboard

Deliverables:

- Score calculation job
- Analytics dashboard

Owner: CDP + AI Agent

Duration: Week 7–8

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4. Development Timeline (Gantt Chart Format)

```
gantt
    title CDP + CRM + WA Automation Development Timeline (P
```

```
arallelized)
    dateFormat  YYYY-MM-DD
    axisFormat  %d %b

    section Phase 0 — Specification
    Architecture Design                    :a1, 2026-03-02,
3d
    Event Schema & Data Model              :a2, after a1, 2
d

    section CDP Development (PIC CDP Dev)
    Customer Profile Service               :b1, 2026-03-04,
5d
    Identity Resolution                    :b2, after b1, 3
d
    Event Ingestion API                    :b3, after b2, 4
d
    Event Store Setup                      :b4, after b3, 3
d
    Point Ledger Core                      :b5, 2026-03-17,
5d
    Event Analytics + Scoring              :b6, 2026-03-27,
5d

    section CRM + Integration (2 Developers)
    Rise CRM Setup                         :c1, 2026-03-04,
4d
    Contact Sync CDP → CRM                 :c2, after c1, 3
d
    Campaign Management Module             :c3, after c2, 3
d
    WABA Integration                       :c4, 2026-03-13,
3d
    Template + Broadcast API               :c5, after c4, 3
d
    WA Flow Integration                    :c6, after c5, 3
d
    Consent + Suppression Layer            :c7, 2026-03-24,
```

```
3d
    Closed Loop Tracking                    :c8, after c7, 3
d

    section Automation Layer (n8n)
    n8n Workflow Infrastructure              :d1, 2026-03-1
3, 3d
    Campaign Automation Flow                :d2, after d1, 3
d
    Retry Queue + Error Handling            :d3, after d2, 2
d

    section AI Agent Development
    AI Agent Architecture                   :e1, 2026-03-04,
4d
    AI Conversation Logic                   :e2, after e1, 5
d
    AI Campaign Assistance Tools            :e3, 2026-03-24,
4d
    AI Optimization Layer                   :e4, after e3, 4
d

    section Final Integration & Stabilization
    System Integration Test                 :f1, 2026-03-30,
4d
    Bug Fix + Performance Tuning            :f2, after f1, 4
d
    Production Readiness                    :f3, after f2, 3
d
```

Development Timeline Table

Week	CDP Dev	AI Agent Dev	CRM + Integration Dev	Output
1	Event schema + profile service	AI agent design	Rise CRM setup	Technical foundation

Week	CDP Dev	AI Agent Dev	CRM + Integration Dev	Output
2	Event ingestion + identity	AI prompt framework	Contact sync	CDP ready
3	Segment engine	Agent integration planning	Campaign module	CRM operational
4	Event processing	AI automation hook	WA integration	Messaging live
5	Ledger design	AI interaction logic	n8n workflow	Automation ready
6	Point ledger build	Agent conversation testing	Consent layer	Loyalty live
7	Event analytics	Agent optimization	Closed-loop tracking	Full automation
8	Scoring engine	AI assistance in campaign	Dashboard + optimization	Production ready

5. MVP Scope (8 Week Target)

MVP Features:

- Customer profile
- Event ingestion
- WA broadcast
- WA Flow survey / quiz
- Campaign management
- Point earn
- Redeem request
- Consent management
- Closed-loop tracking

Not required for MVP:

- Advanced AI recommendation

- Advanced scoring models
- Multi-channel orchestration

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6. Risk Mitigation

Major risks:

1. Identity duplication
2. Message delivery failure
3. Data inconsistency
4. Point ledger corruption

Mitigation:

- Identity resolution rules
- Retry queue in n8n
- Immutable ledger
- Event validation

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7. Expected System State After 8 Weeks

Platform sudah mampu:

- Mengelola customer profile
- Melakukan segmentasi
- Menjalankan campaign WhatsApp
- Menjalankan survey / quiz / task
- Mengelola point loyalty
- Melakukan redeem
- Menjalankan automation
- Melakukan analytics dasar

Platform siap masuk fase:

- Growth experimentation

- Advanced scoring
- AI personalization